IIMT3642: Managing and Mining Big Data - Clustering

Zhepeng (Lionel) Li, PhD
Associate Professor
Innovation and Information Management

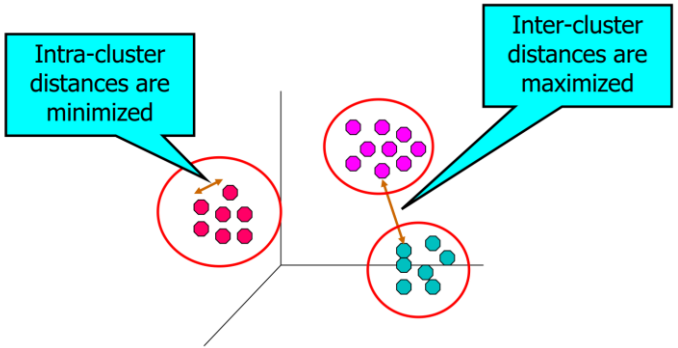
Lecture Outline

1. What is clustering
2. Measure - Distance
3. Clustering Methods
   - K-means Clustering
   - Hierarchical Clustering
   - DBSCAN

Readings: Chapter 8 in Data Mining Concepts and Techniques 4th Edition

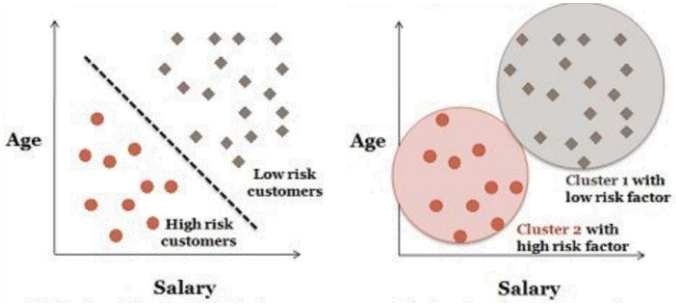
What is clustering?

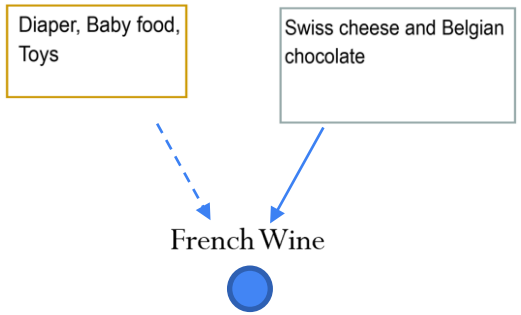
- Clustering: group objects based on the data describing them
  - Objective: objects within a group similar to each other and different from objects in other groups.
    - Performance measuring - The greater the similarity within a group and the greater the difference between groups, the better the clustering.



What is not clustering?

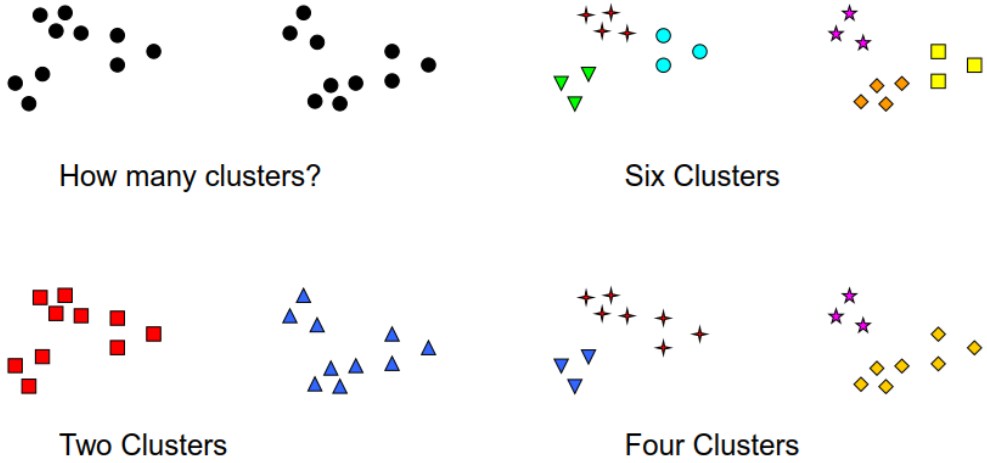
- Simple segmentation
  - Divides objects based on an external specification or predefined rule (e.g., dividing students into registration groups alphabetically, by their last name)
- Classification
  - Learns a model for supervised information (i.e., class label)
- Association Rules Mining
  - Local vs. global connections





Notion of a cluster can be ambiguous

- Clustering is an unsupervised method where each sample is without class label
  - Clustering is not meant to be unique or approaching the “ground truth”



Measure - Distance

- For two objects X and Y with ***n*** continuous attributes (e.g., numerical attributes), the Euclidean distance between them is defined as:

$$d(X,Y) = \sqrt[2]{(x_1-y_1)^2+(x_2-y_2)^2+\ldots+(x_n-y_n)^2}$$

In which, xs and ys are attribute values of object X and Y, respectively

- Distance between two objects reflect how dissimilar these two objects are

Example: Euclidean distance

Name	Age	Balance ($)
Sue	21	23,200
Carl	27	23,330
Tom	45	23,260
Jack	52	23,400

Potential issue: unit difference

Normalization

- To alleviate the effect of measurement unit on distance calculation, attributes are normalized or standardized before distance calculation.
  - Map every attribute value to be within [0,1] range, by applying normalization options:
    - for example, Minmax normalization, where we convert the values as

$$NewValue = \frac{(OldValue - MIN)}{(MAX - MIN)}$$

In which, the MAX and MIN are the maximal and minimal values of the given attribute, respectively

K-means Clustering

- Assign ***n*** objects to ***K*** groups such that Sum of the Squared Errors (SSE) defined below is minimized.

$$SSE = \sum_{i=1}^{K} \sum_{x \in C_i} [d(mean_i, x)]^2$$

In which,

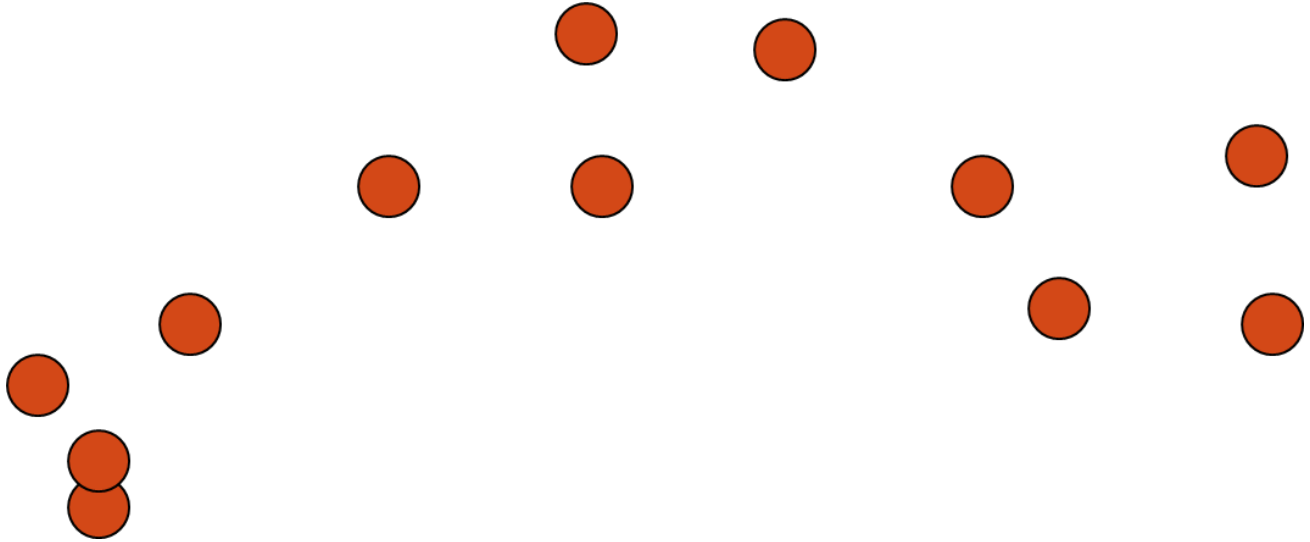
  - ***K*** groups: $C_1, C_2, \ldots, C_K$
  - $Mean_i$: mean of the group C_i, $i = 1, 2, \ldots, K$

- Computational Challenge
  - There are about K^n different ways of assigning n objects to K groups

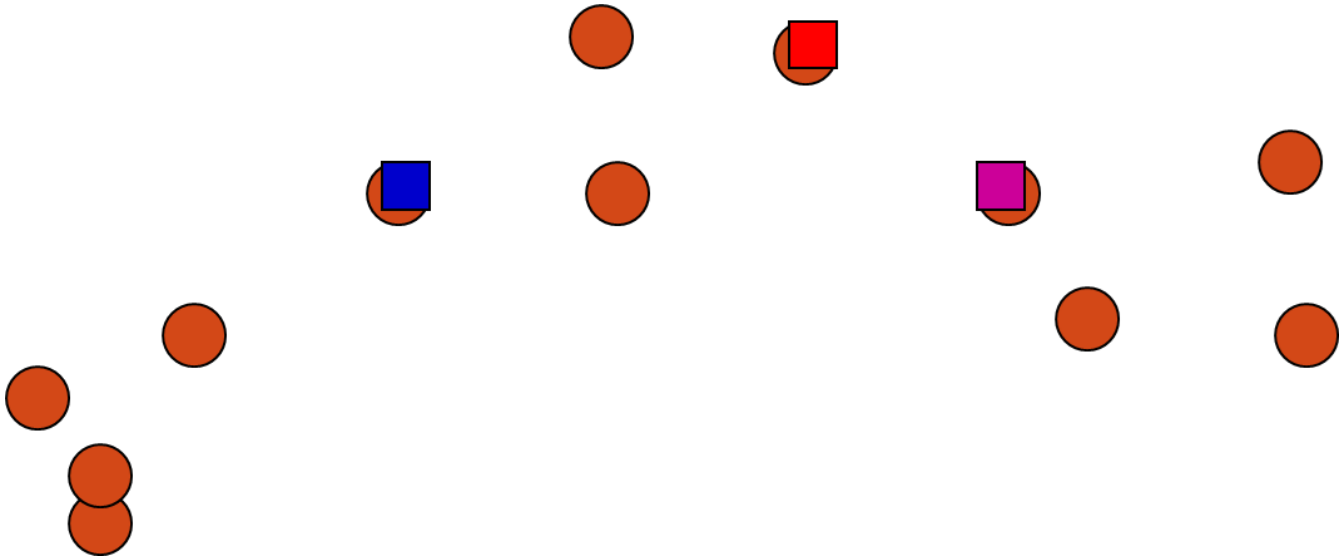
- K-means algorithm:
  - Step 1: randomly select K objects as initial means (aka. centroids)
  - Step 2: form K clusters by assigning each object to its closest mean
  - Step 3: Re-compute the mean of each cluster
  - Step 4: Go to Step 2 until means do not change

Example: executing K-means algorithm

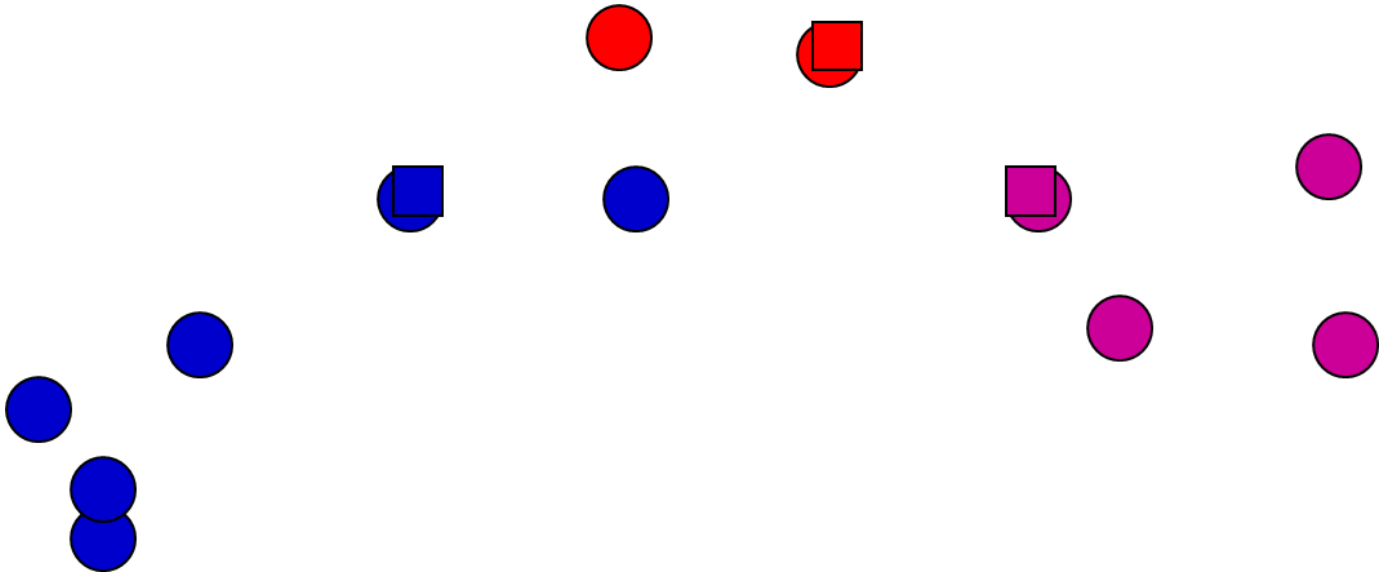
Input Data



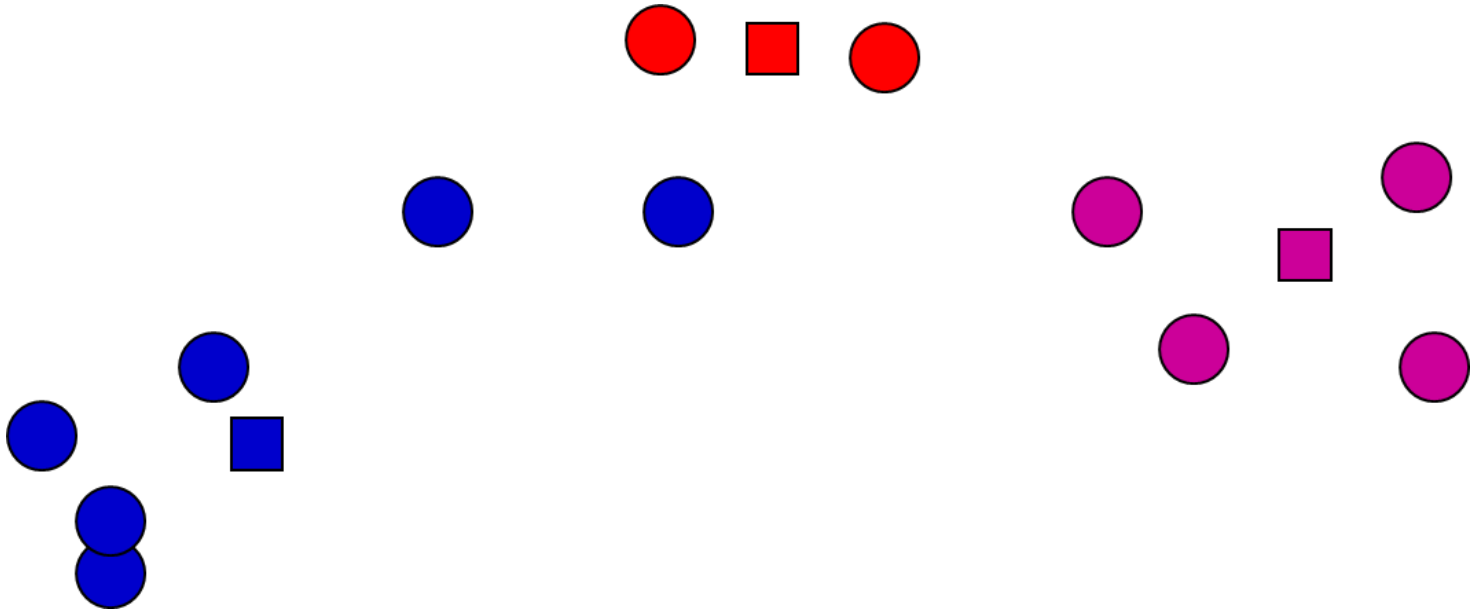
randomly select K objects as initial means



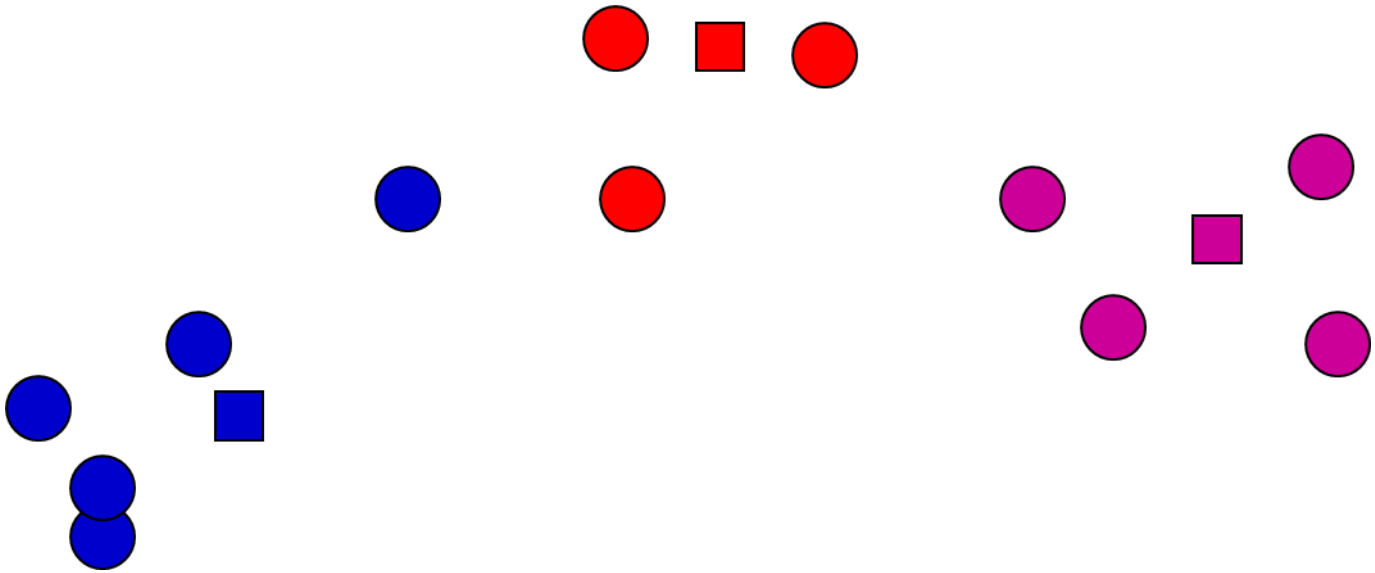
assign each object to its closest mean



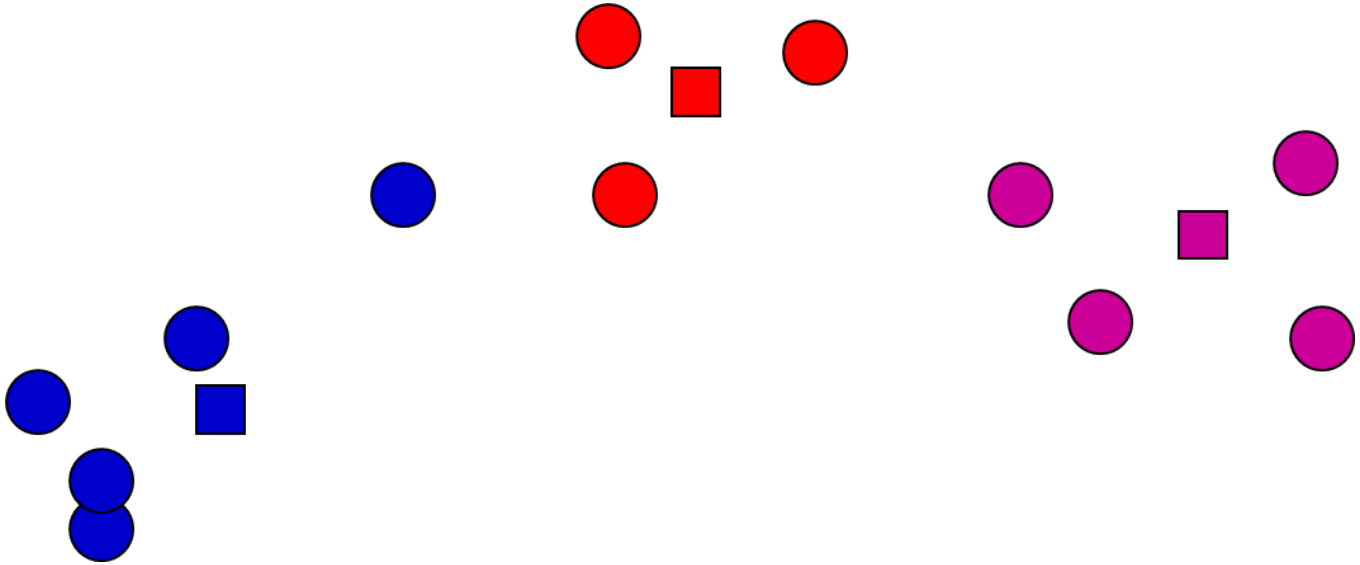
re-compute mean of each cluster



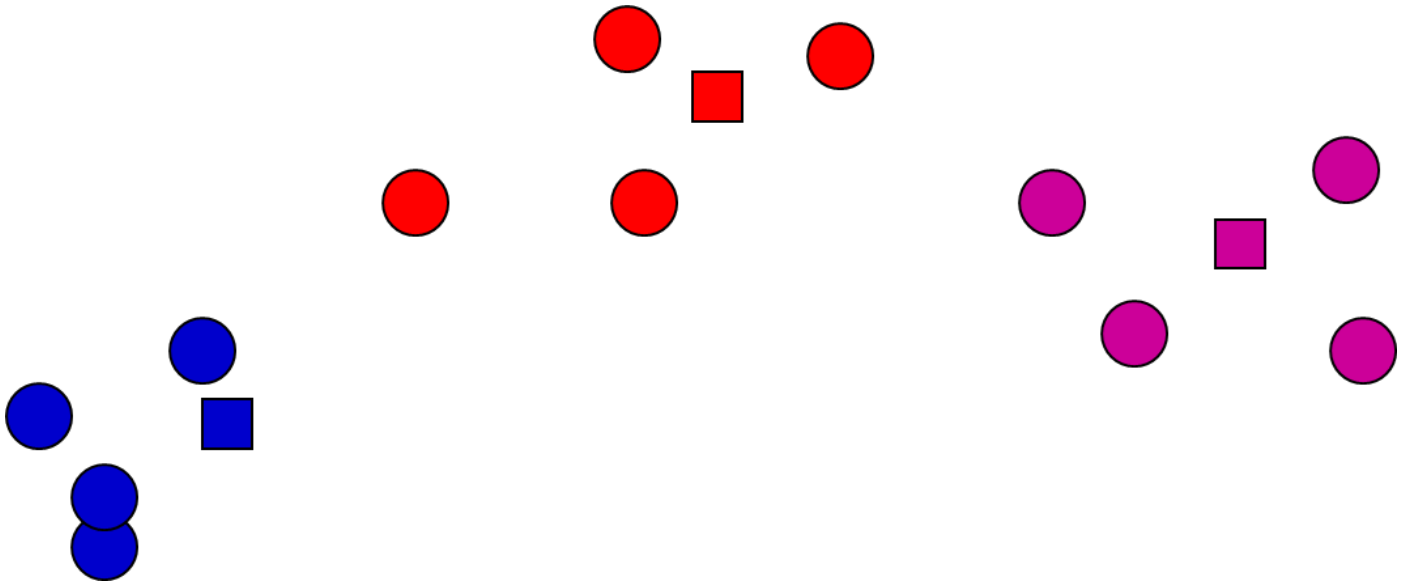
assign each object to its closest mean



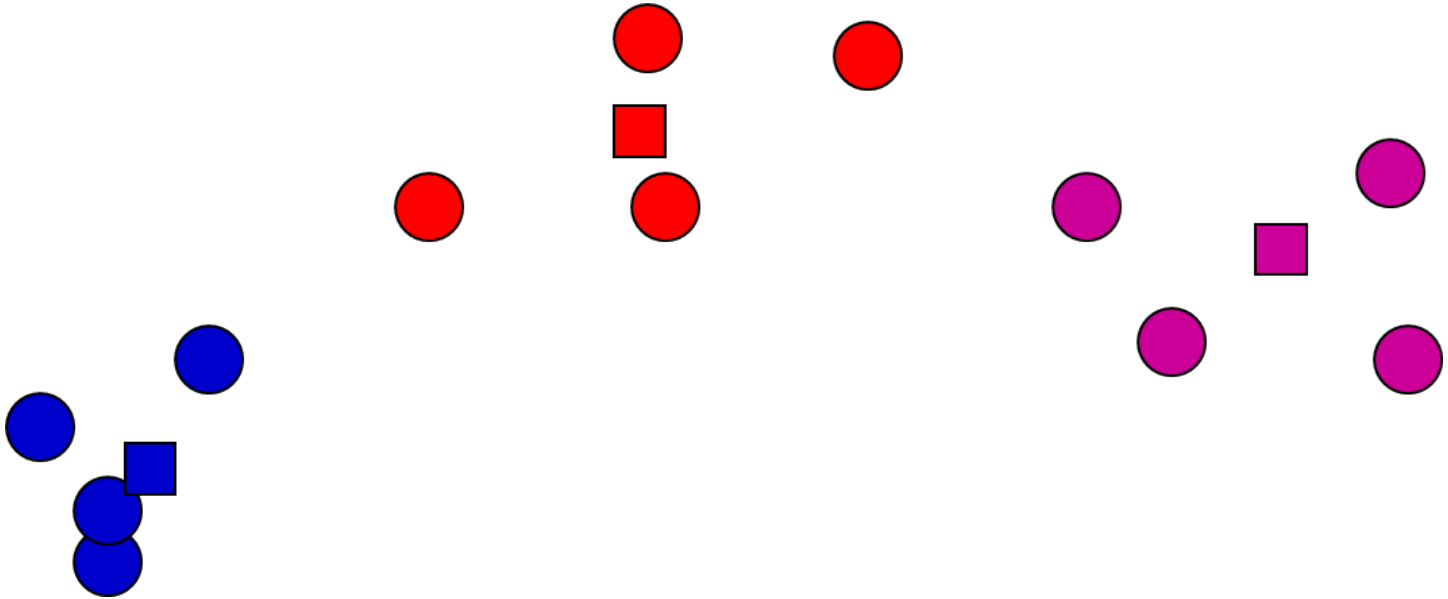
re-compute mean of each cluster



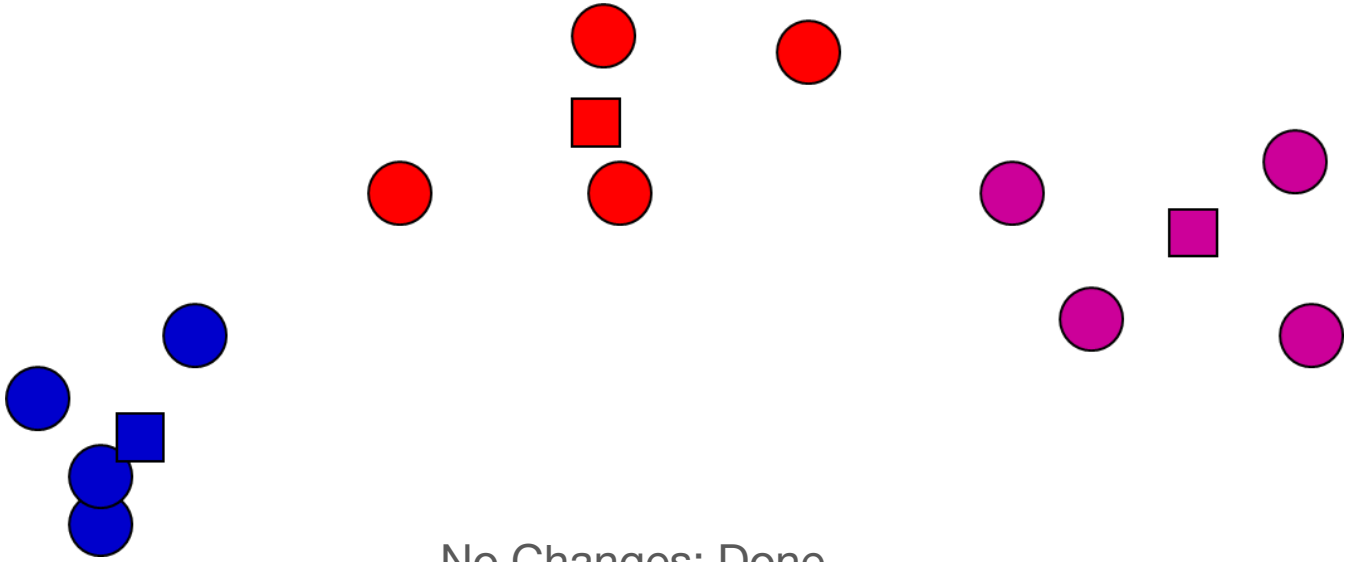
assign each object to its closest mean



recompute mean of each cluster



assign each object to its closest mean



No Changes: Done

An online demo for K-means clustering

https://user.ceng.metu.edu.tr/~akifakkus/courses/ceng574/k-means/

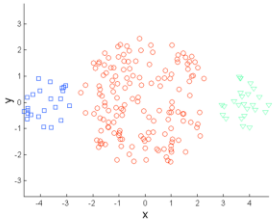
K-means clustering

- Strengths
  - A classical clustering algorithm with good performance
  - Simple: easy to implement
  - Efficient: fast in terms of computational time - $O(N)$ in complexity of each update
- Weaknesses
  - Only applicable to objects with numerical attributes
  - No substructure inside the group (e.g., hierarchies)
  - The best number of K is to be determined

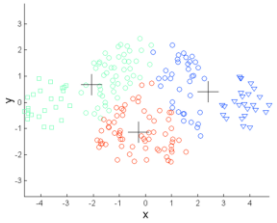
Determining K

- Clustering results heavily depend on initialization (Demo)
  - Location?
  - Number?

Limitations of K-means: Differing Sizes

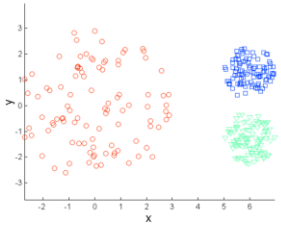


Original Points

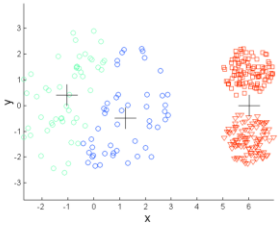


K-means (3 Clusters)

Limitations of K-means: Differing Density

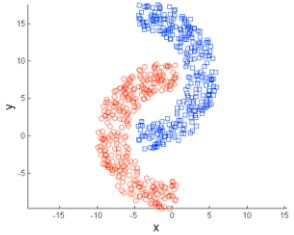


Original Points

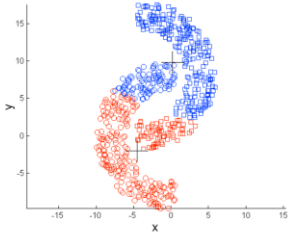


K-means (3 Clusters)

Limitations of K-means: Non-globular Shapes



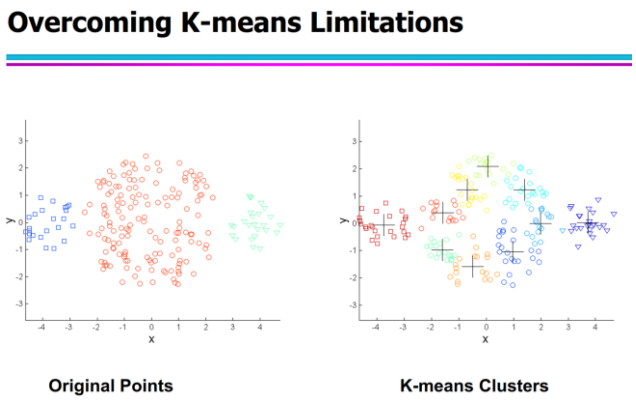
Original Points

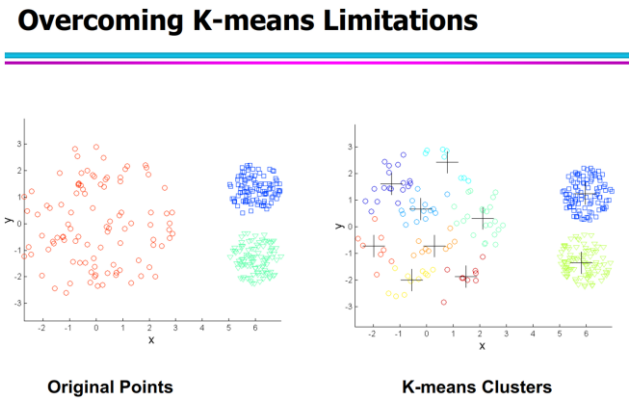


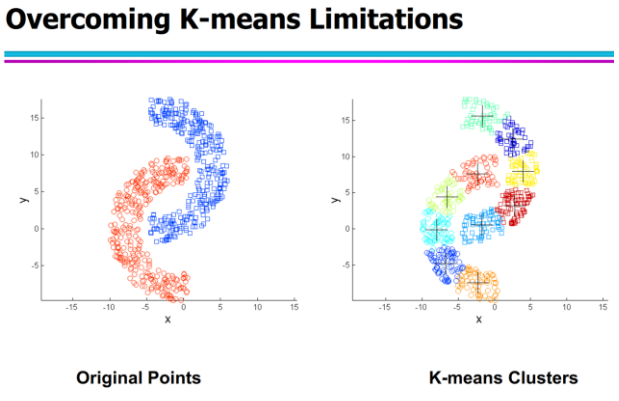
K-means (2 Clusters)

Determining K

- Clustering results heavily depend on initialization (Demo)
  - One option is to use many clusters, and decide how to combine them later
    - K-means will approach the location of centroids when the number is appropriate.

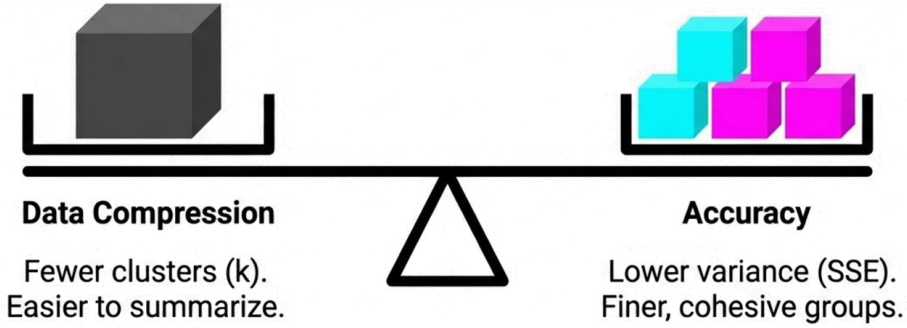






Determining K

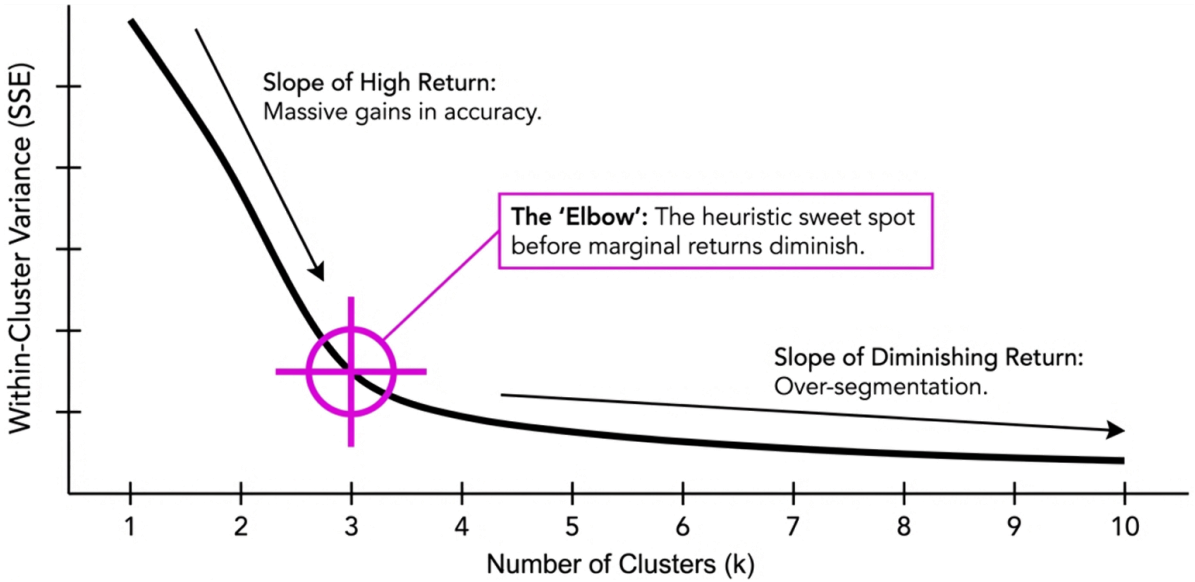
- The Elbow Method
  - Intuition: the trade-off by increasing K



Increasing k naturally reduces within-cluster variance. However, splitting an already cohesive cluster only yields a tiny marginal reduction in error.

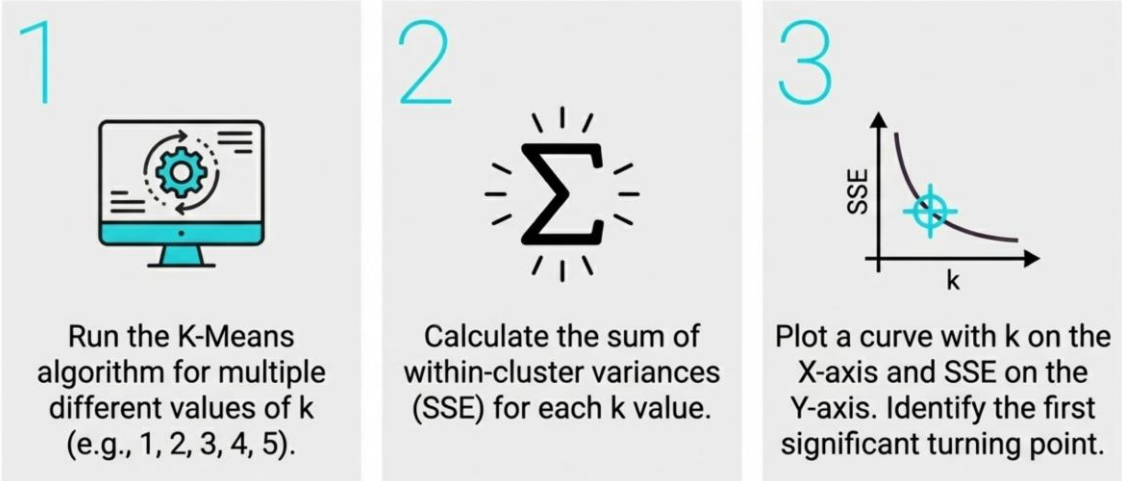
Determining K

- The Elbow Method
  - Finding the right number of clusters (i.e., K)



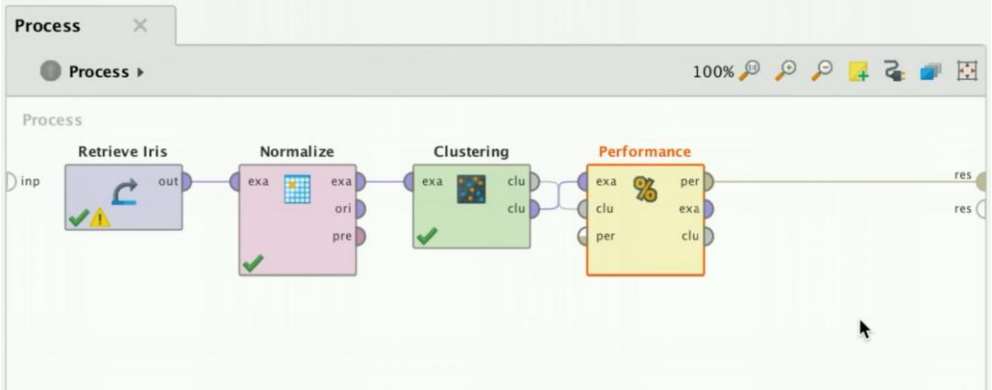
Determining K

- The Elbow Method
  - Executing in actionable steps



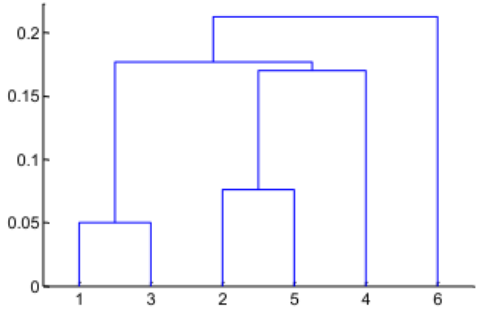
Determining K

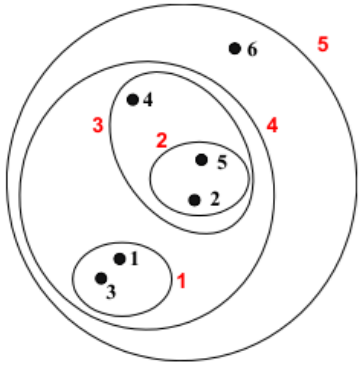
- The Elbow Method
  - Executing in practice (e.g., SSE can be calculated by “Performance” under the Evaluation operators in RapidMiner)



Hierarchical Clustering

- Address the lack of substructure for K-means
- Produces a set of nested clusters organized as a hierarchical tree
- Visualized as a dendrogram

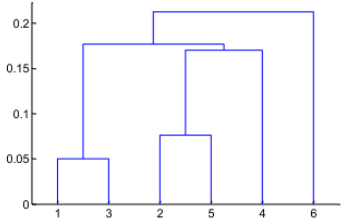


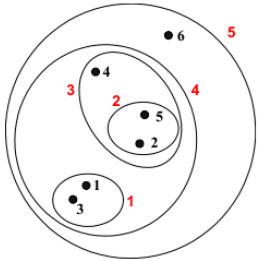


Hierarchical Clustering

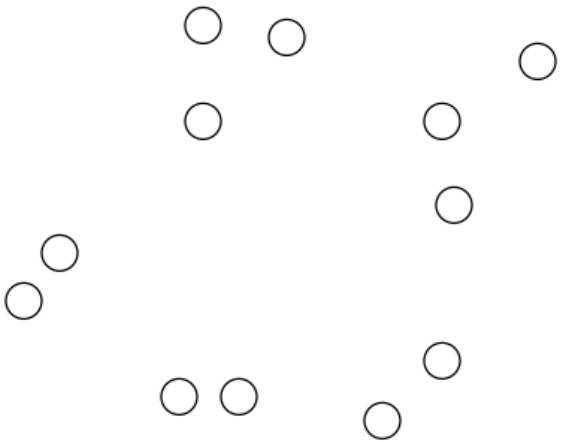
Agglomerative Clustering:

- Step 1: Place each object in an unique cluster.
- Step 2: Calculate distance between clusters.
- Step 3: Merge two clusters with the least distance.
- Step 4: Go to step 2 until all objects are in one cluster.





Start with clusters of individual points and a proximity matrix



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
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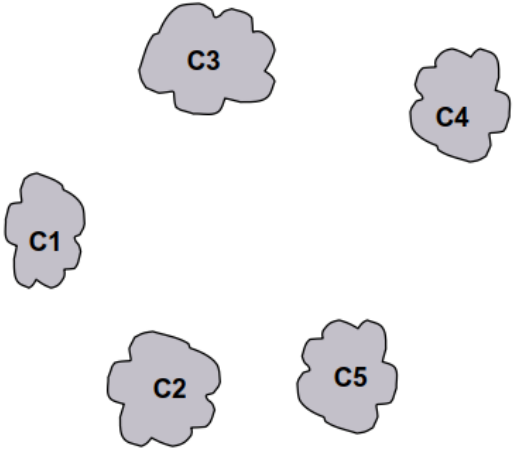
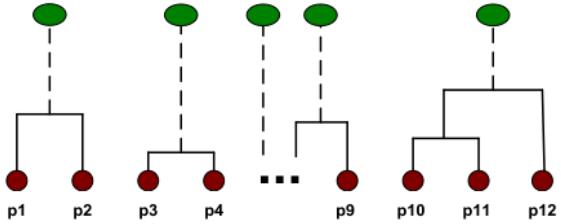
Proximity Matrix

After merging steps, we will have clusters

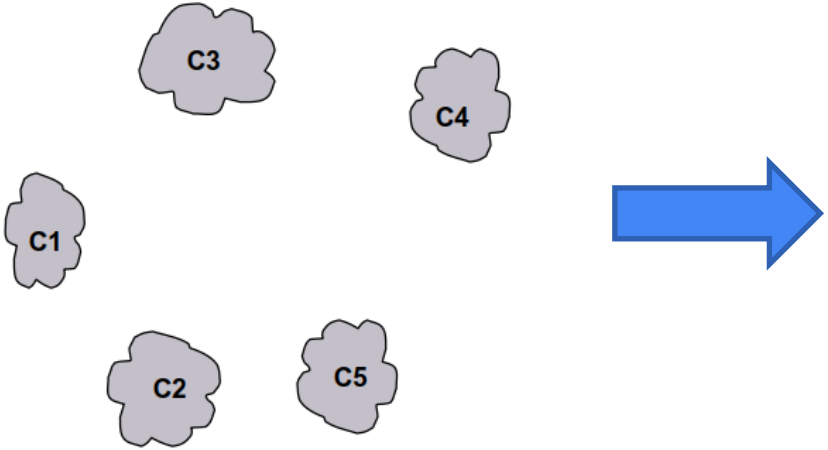


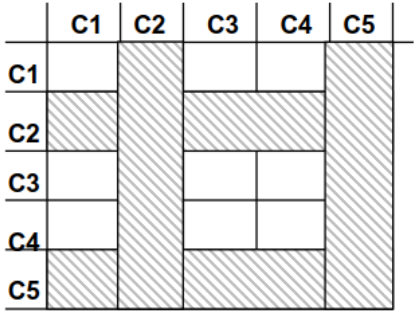
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix

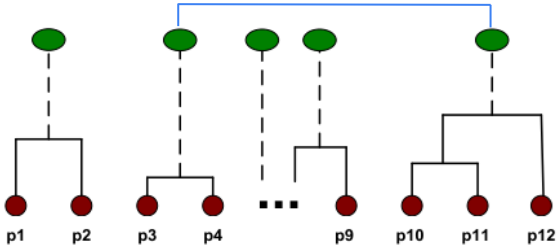


We want to merge the two closest clusters (say C2 and C5) and update the proximity matrix

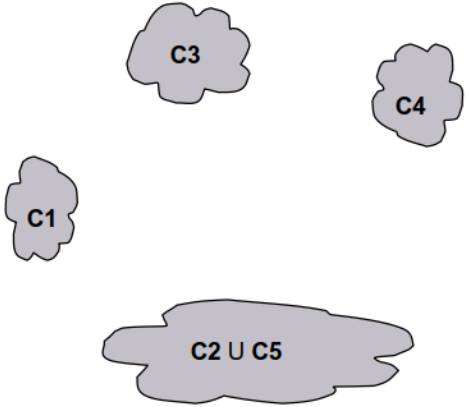




Proximity Matrix

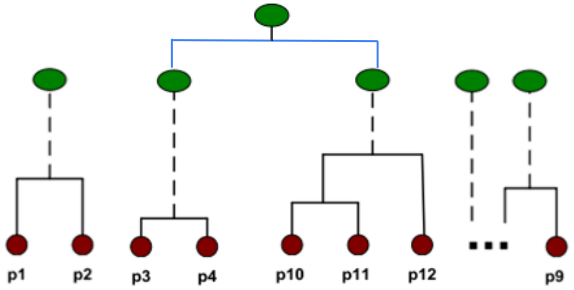


After Merging, the question is “How do we update the proximity matrix”



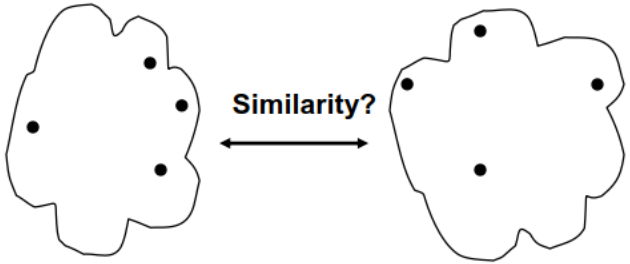
	C1	C2 U C5	C3	C4
C1		?		
C2 U C5	?	?	?	?
C3		?		
C4		?		

Proximity Matrix



Inter-cluster Distances

- MIN
- MAX
- Group Average



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
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Proximity Matrix

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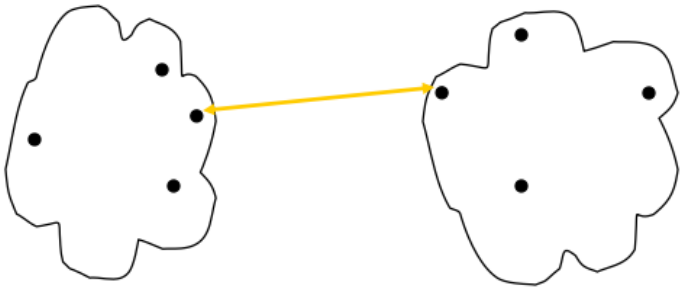
- MAX
- Group Average



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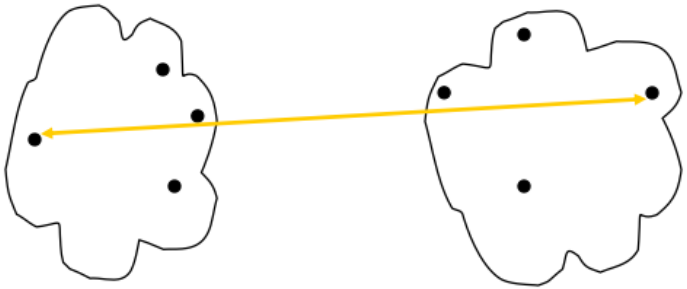
Proximity Matrix

Inter-cluster Distances

- MIN
- MAX
- Group Average



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p1						
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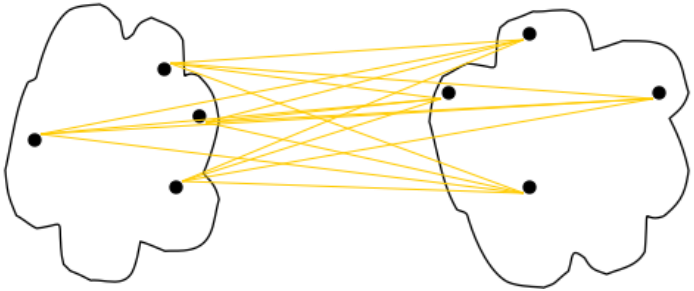
Proximity Matrix

Inter-cluster Distances

- MIN
- MAX
- Group Average



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

Example: executing hierarchical clustering

	Sue	Tom	Carl	Jack	Mary
Sue	0	6	8	2	7
Tom	6	0	1	5	3
Carl	8	1	0	10	9
Jack	2	5	10	0	4
Mary	7	3	9	4	0

Proximity Matrix

M S J T C

Dendrogram

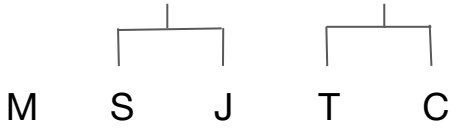
Group Average

	Sue	T, C	Jack	Mary
Sue	0	7	2	7
T,C	7	0	7.5	6
Jack	2	7.5	0	4
Mary	7	6	4	0

Dendrogram

	S,J	T, C	Mary
S,J	0	7.25	5.5
T,C	7.25	0	6
Mary	5.5	6	0

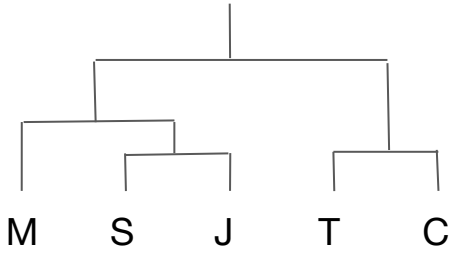
	Sue	Tom	Carl	Jack	Mary
Sue	0	6	8	2	7
Tom	6	0	1	5	3
Carl	8	1	0	10	9
Jack	2	5	10	0	4
Mary	7	3	9	4	0



Dendrogram

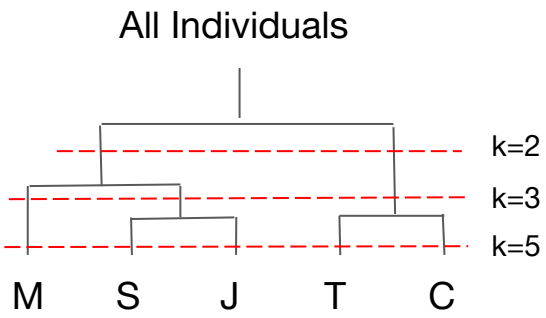
	S,J	T, C	Mary
S,J	0	7.25	5.5
T,C	7.25	0	6
Mary	5.5	6	0

	Sue	Tom	Carl	Jack	Mary
Sue	0	6	8	2	7
Tom	6	0	1	5	3
Carl	8	1	0	10	9
Jack	2	5	10	0	4
Mary	7	3	9	4	0



Dendrogram

- Strengths:
  - No need to assume particular number of clusters (desired number of clusters can be obtained by cutting the dendrogram at proper level)
  - Studies suggest that it produces meaningful taxonomies (e.g., animal kingdom)
- Weaknesses:
  - Computational expensive
  - All the merges are final

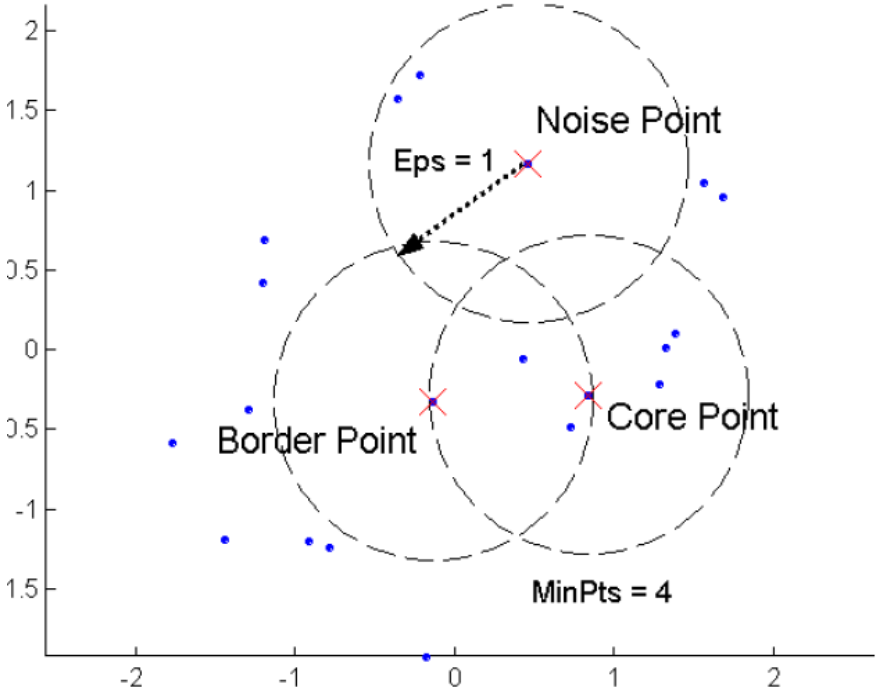


Dendrogram

DBSCAN

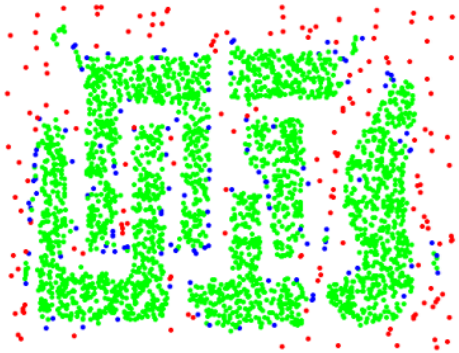
- Density-based method that addresses the issue of differing density and non-globular shape in the data points
  - **Epsilon (ϵ):** the radius of a neighborhood centered around a given object
  - **MinPts:** the density threshold, specifying the minimum number of objects that must exist within that ϵ-neighborhood
  - Every data point is categorized into one of the three types:
    - Core: An object is a core object if its ϵ-neighborhood contains at least MinPts objects
    - Border: Objects that are not core objects themselves, but they fall inside the ϵ-neighborhood of a core object
    - Noise (outliers): Any object that does not belong to any ϵ-neighborhood of a core object is simply marked as noise or an outlier

Example: Core, Border, and Noise Objects



Example: Core, Border, and Noise Objects

Original Points



Point types: core, border and noise

Eps = 10, MinPts = 4

DBSCAN algorithm

- Step 1: The algorithm randomly selects an unvisited object
- Step 2: If it has < MinPts in its neighborhood, it is temporarily marked as noise
- Step 3: If it has >= MinPts, a new cluster is created, and all objects in its neighborhood are added to this cluster
- Step 4: The algorithm iteratively checks the newly added objects; if they are also core objects, their neighborhoods are added to the cluster too, continuing until the dense region cannot be expanded any further

An online demo for DBSCAN

https://www.naftaliharris.com/blog/visualizing-dbscan-clustering/